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Please fill in answers in the space supplied and show all your work to receive full credit.
Useful Tables at end of Exam.

1. a) (10 pts) State whether the following statements are **True or False**
- An ideal gas at a given state expands to a fixed final volume at constant pressure or at constant temperature. Work is greater in constant temperature case.
 - A two-phase liquid-vapor mixture with equal volumes of saturated vapor and saturated liquid has a quality of 0.5.
 - As pressure decreases from the critical pressure, the values of the specific volume of saturated vapor and saturated liquid approach each other.
 - If a system undergoes a polytropic process, the system must consist of an ideal gas.
 - It takes more energy to vaporize 1kg of saturated liquid water at 100°C than it would at 120°C.
- b) (5 pts) The gage pressure of an automobile tire is measured to be 210kPa before a trip and 220kPa after the trip at a location where the atmospheric pressure is 95kPa. Assuming the volume of the tire remains constant and the air temperature before the trip is 25°C, determine the air temperature in the tire after the trip. (Air is ideal gas)
- c) (5pts) The specific volume of saturated liquid water at 0.01°C is $1.002 \times 10^{-3} \text{ m}^3/\text{kg}$. What would be the specific volume of saturated solid at the same temperature, greater or less than saturated liquid value? Why?

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2. (50 pts) 2kg of liquid-vapor mixture of water contained in a piston–cylinder assembly undergoes two processes in series from an initial state where the temperature is 120°C.

- **Process 1–2** System is heated at constant volume to a saturated vapor state at 200°C.
- **Process 2–3** System is compressed at constant pressure until it reaches to 150°C.

Kinetic and potential effects can be ignored.

- a) (3 pts) Draw a schematic of the system and label the system boundary.
- b) (8 pts) Carefully sketch the process on a:
 - (1) P-v diagram (Pressure-specific volume)
 - (2) T-v diagram (Temperature-specific volume)
- c) (5 pts) State the engineering model
- d) (2 pts) Write the simplified energy balance
- e) (15 pts) Calculate the work transferred in kJ. Is work done on the system or by the system?
- f) (15 pts) Calculate the heat transferred in kJ. Is heat transferred to the system or from the system?
- g) (2 pts) Calculate the mass of vapor in each state in kg.

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3. (30 points) A piston cylinder device initially contains 0.4m^3 of air at 100kPa and 240°C . The air is now compressed to 0.1m^3 in such a way that the temperature inside the cylinder remains constant. The air is modeled as an ideal gas and kinetic and potential energy effects are negligible.
- a) (2 pts) Draw a schematic of the system and label the system boundary.
 - b) (2 pts) State the engineering model
 - c) (3 pts) Write the simplified energy balance and compressibility factor (Z).
 - d) (3pts) Draw P-v diagram (Pressure-specific volume)
 - e) (10 pts) Determine the work (kJ)
 - f) (10 pts) Determine the heat (kJ)

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